PATENT SPECIFICATION

System and Method for Dynamic Transaction Flow Control in Financial Networks Using Fluid Dynamics Modelling

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1. TECHNICAL FIELD

The present invention relates to the field of computer-implemented control systems for financial transaction processing networks, and more particularly to systems and methods for dynamically monitoring systemic risk and computing state-adaptive transaction-flow control actions using mathematical models drawn from classical mechanics, fluid dynamics, and spectral graph theory.

2. BACKGROUND OF THE INVENTION

2.1. Technical Problem

Transaction processing networks, including interbank payment systems and distributed ledger transaction networks, are tasked with processing large volumes of value-transfer events under adversarial and rapidly changing conditions. Operators of such systems (including protocol operators, compliance gateways, and transaction validators) require mechanisms to dynamically control transaction flow in response to systemic risk conditions. Current approaches suffer from several fundamental limitations:

(a) **Static enforcement parameters:** Existing systems often apply fixed risk thresholds, fixed transaction rate limits, or fixed escrow rules regardless of the structural state of the underlying transaction network.

(b) **No formal phase transition detection:** Existing approaches do not identify the critical threshold above which constant controls necessarily fail and state-adaptive controls become essential.

(c) **Retroactive indicators:** Many systemic risk measures are contemporaneous or lagging indicators. They confirm that instability is occurring or has occurred, rather than providing advance warning usable for real-time transaction flow control.

(d) **Absence of interaction modelling:** Standard approaches treat institutions or addresses independently or through pairwise conditional measures. They do not model the N-body interaction dynamics that give rise to emergent systemic phenomena such as herding, contagion cascades, and collective phase transitions.

(e) **No closed-form optimal controller:** While optimal control theory is well-developed in engineering, no prior work derives a closed-form expression for the optimal state-adaptive friction (flow control) as a function of the financial network state for use in transaction processing systems.

2.2. Prior Art Limitations

Prior art known to the applicant includes:

- **CoVaR** (Adrian and Brunnermeier): Measures the Value-at-Risk of the financial system conditional on one institution being in distress. This is a pairwise conditional measure and does not model N-body interactions.
- **SRISK** (Brownlees and Engle): Measures the expected capital shortfall of an institution in a systemic event. This is an institution-level measure and does not derive regulatory policy.
- **Basel III CCyB rule:** Maps the credit-to-GDP gap to a buffer using a fixed piecewise-linear function. This is a constant-friction policy and does not adapt to the structural state of the network.
- **Agent-based models:** Simulate financial networks using behavioural rules. While capable of generating complex dynamics, they typically lack analytical gradient flow, provable stability guarantees, and closed-form optimal policies.
- **Network contagion models** (Acemoglu, Ozdaglar, and Tahbaz-Salehi): Model contagion on known network topologies using equilibrium analysis. They do not model dynamic energy landscapes or derive spectral phase transition criteria from the interaction matrix.

3. SUMMARY OF THE INVENTION

The present invention provides a system and method for monitoring systemic risk and computing state-adaptive transaction-flow control actions in real time. The key contributions are:

(i) A **gravitational interaction engine** (GravityEngine) that models N financial institutions as particles in $\mathbb{R}^d$ interacting through a pairwise potential comprising erf-attraction and logarithmic repulsion, with mean-reversion and blind-spot energy coupling.

(ii) A **critical manifold** $\mathcal{C}$ defined by the condition $\lambda_{\max}(\rho(X) \cdot \ell(X) \cdot W) = 1$, above which constant friction provably fails and state-adaptive regulation is necessary.

(iii) A **closed-form optimal adaptive policy** $\gamma^*(X)$ derived from the Bellman equation, with the property that $\gamma^* \to 0$ when systemic energy approaches zero.

(iv) An **equivalence theorem** proving that the blind-spot energy and the gravitational potential are equivalent Lyapunov functions, providing dual perspectives on systemic stability.

(v) A **Navier–Stokes fluid dynamics analogy** computing vorticity, enstrophy, and blow-up criteria from the financial network state for regime classification (hedge, speculative, Ponzi).

(vi) **CCyB calibration** mapping the adaptive friction coefficient to countercyclical capital buffer recommendations in basis points.

(vii) A **transaction-flow control mapping** that maps the adaptive friction coefficient to one or more enforcement actions comprising one or more of: risk threshold selection, transaction rate limiting, queue delay, escrow duration control, maximum value limits, and additional verification requirements for high-risk conditions.

4. DETAILED DESCRIPTION OF THE INVENTION

4.1. GravityEngine: Multi-Body Financial Dynamics

The system models N financial institutions as particles in a d-dimensional feature space, where the feature vector of each institution encodes financial metrics including leverage ratio, asset size, interconnectedness, and risk-weighted capital adequacy. The dynamics are governed by:

$$\dot{X} = -\alpha(X - \mu\mathbf{1}^\top) - \nabla_X \Phi_{\text{pair}}(X) - \sum_k \beta_k D\Phi_k(X)^\top D_k(X) \tag{1}$$

where:

- $X \in \mathbb{R}^{N\times d}$ is the state matrix (rows are institutions, columns are features);
- α is a mean-reversion rate pulling institutions toward the system centroid μ;
- Φ_{pair} is the pairwise interaction potential;
- Φ_k are external field operators including blind-spot energy channels; and
- β_k are coupling strengths for each external field.

4.1.1 Pairwise Interaction Potential

The pairwise interaction between institutions i and j is governed by a potential comprising erf-attraction and logarithmic repulsion:

$$\phi(r) = \gamma \cdot \frac{\sigma\sqrt{\pi}}{2} \cdot \operatorname{erf}\left(\frac{r}{\sigma}\right) - \gamma\lambda_{\text{rep}} \cdot \log(r + \varepsilon) \tag{2}$$

where $r = \|x_i - x_j\|$ is the inter-institution distance, γ is the interaction strength, σ the interaction scale, λ_{rep} the repulsion strength, and ε a regularisation constant preventing divergence at zero distance.

The erf-attraction provides a smooth, bounded attractive force that saturates at large distances (unlike gravitational $1/r$ attraction which is unbounded), while the logarithmic repulsion prevents institution collapse—analogous to the Pauli exclusion principle in quantum mechanics.

The equilibrium distance at which attraction and repulsion balance is:

$$r^* = \sigma\sqrt{\log(1/\lambda_{\mathrm{rep}})} \tag{3}$$

In the preferred embodiment: $\alpha = 0.10$, $\gamma = 1.00$, $\sigma = 1.00$, $\lambda_{\mathrm{rep}} = 0.10$, $\varepsilon = 10^{-5}$, with force clamping at $F_{\max} = 100.0$.

4.2. Blind-Spot Energy Module

The system incorporates a blind-spot energy function computed from four deviation channels, referred to herein by the Blind Spot Decomposition Theory (BSDT) framework:

(a) **Enstrophy anomaly channel (δ_C):** Measures deviation of the system's enstrophy (rotational energy) from the reference period.

(b) **Spectral gap channel (δ_G):** Measures deviation of the spectral gap of the correlation matrix from reference levels, indicating structural fragility.

(c) **Alignment anomaly channel (δ_A):** Measures deviation of the co-movement alignment of institutions from reference patterns.

(d) **Temporal novelty channel (δ_T):** Measures the degree to which the current system state is temporally novel relative to the reference period.

The blind-spot energy is computed as:

$$E_{\mathrm{BS}}(X) = \|(X-\mu_0)^\top \Sigma_0^{-1}(X-\mu_0)\|_F \tag{4}$$

where μ_0 and Σ_0 are the mean and covariance of the reference (normal) period, computed using Ledoit–Wolf Oracle shrinkage estimation.

The gradient of the blind-spot energy provides the BSDT score:

$$\nabla E_{\mathrm{BS}} = 2\Sigma_0^{-1}(X-\mu_0) \tag{5}$$

4.3. Equivalence Theorem

A key theoretical result establishes the equivalence of the blind-spot energy and the gravitational potential as stability indicators:

Theorem C (Lyapunov Equivalence): The blind-spot energy E_{BS} and the GravityEngine potential Φ are equivalent Lyapunov functions in the sense that there exist constants $c_1, c_2 > 0$ such that:

$$c_1\|\nabla\Phi\| \le \|\nabla E_{\mathrm{BS}}\| \le c_2\|\nabla\Phi\| \tag{6}$$

Empirically, the cosine similarity between the two gradients is $\cos\theta = -0.335$, confirming that the two energy functions are anti-aligned: they measure *complementary* aspects of systemic instability. The gravitational potential captures interaction-driven instability, while the blind-spot energy captures drift-driven instability invisible to equilibrium measures.

4.4. Critical Manifold and Phase Transition

The system computes a critical manifold $\mathcal{C}$ in the state space that demarcates the phase transition between regimes where constant friction suffices and regimes where it necessarily fails:

Theorem 2 (Spectral Phase Transition):

$$\mathcal{C} = \{X : \lambda_{\max}(\rho(X) \cdot \ell(X) \cdot W) = 1\} \tag{7}$$

where:

- $\rho(X)$ is the spectral radius of the inter-institution correlation matrix at state X;
- $\ell(X)$ is a leverage factor derived from the system leverage; and
- W is the weighted adjacency matrix of the interaction network.

Below $\mathcal{C}$: the system is in a sub-critical regime where constant flow-control parameters provide adequate damping.

Above $\mathcal{C}$: the system is in a super-critical regime where any constant friction coefficient is provably suboptimal. State-adaptive friction is necessary and sufficient for stability.

The spectral radius is computed using power iteration with finite-difference Hessian-vector products ($h = 10^{-4}$), achieving $O(N^2 d \log N)$ computational complexity per time step.

4.5. Optimal Adaptive Policy

Theorem OR2 (Closed-Form Optimal Friction): The optimal state-dependent friction coefficient solving the associated Bellman equation is:

$$\gamma^*(X) \approx \frac{\beta \|\nabla E\|^2}{2\kappa + \beta \|\nabla E\|^2} \cdot \frac{E(X)}{E(X) + \theta} \tag{8}$$

where:

- $E(X)$ is the combined systemic energy (gravitational + blind-spot);
- $\|\nabla E\|$ is the gradient norm measuring the rate of energy change;
- κ is the economic cost parameter of imposing friction;
- β is the systemic damage coefficient; and
- $\theta = \kappa/\beta$ is a damping threshold.

This policy has the desirable property that $\gamma^*(X) \to 0$ as $E(X) \to 0$: no regulatory friction is imposed when the system is stable. Conversely, $\gamma^*(X) \to 1$ as $E(X) \to \infty$: maximum friction is applied during severe instability.

Theorem OR3 (Necessity of State-Dependence): For any constant friction $\bar{\gamma}$, the welfare gap $V^*(X) - V_{\bar{\gamma}}(X) > 0$ whenever the system state X lies above the critical manifold $\mathcal{C}$.

4.6. Transaction Flow Control Mapping

The adaptive friction coefficient is mapped to one or more technical transaction-flow control actions in a transaction processing network. In a preferred embodiment, the mapping produces one or more of:

(a) a selected risk threshold profile for transaction classification;

(b) a transaction rate limit for high-risk counterparties or regions of the transaction graph;

(c) a queue delay or priority adjustment for transactions requiring review;

(d) an escrow duration or lock-up parameter for escrowed transactions;

(e) a maximum allowed transaction value for designated categories; and

(f) additional verification requirements, including requiring proofs or attestations during stressed or critical regimes.

The mapping may be implemented as one or more monotone functions of $\gamma^*(X)$ and/or discrete regime selection based on whether the system state lies above or below the critical manifold $\mathcal{C}$.

4.6.1 Regulatory Mapping (Optional)

In an optional embodiment, the adaptive friction coefficient is also mapped to a countercyclical capital buffer recommendation:

$$\Delta\text{CCyB}(t) \approx \kappa^{-1} \cdot \gamma^*(X_t) \cdot \sigma_\ell(t) \tag{9}$$

where $\sigma_\ell(t)$ is the system leverage volatility. The output is in basis points and represents the recommended adjustment to the countercyclical buffer at time t.

4.7. Welfare Cost of Inaction

The system computes the welfare cost of not implementing adaptive friction:

$$\%CL = 100 \times \left(1 - e^{-\mathcal{L}_{\text{inaction}}/(\sigma\theta T)}\right) \tag{10}$$

where $\mathcal{L}_{\text{inaction}}$ is the accumulated loss under no intervention, σ is a scaling constant, and T is the time horizon.

4.8. Navier–Stokes Fluid Dynamics Analogy

In an enhanced embodiment, the system implements a full fluid dynamics analogy that provides additional regime classification capabilities:

4.8.1 Velocity Field and Strain

The matrix $\dot{X}$ of institution position changes is treated as a velocity field. The velocity gradient tensor is decomposed into symmetric (strain rate) and antisymmetric (vorticity) components:

$$\nabla v = \frac{1}{2}(\nabla v + (\nabla v)^\top) + \frac{1}{2}(\nabla v - (\nabla v)^\top) = S + \Omega \tag{11}$$

4.8.2 Vorticity and Enstrophy

The vorticity tensor $\omega = \nabla v - (\nabla v)^\top$ measures rotational flows in the financial system (characteristic of herding and feedback loops). The enstrophy:

$$\mathcal{E} = \frac{1}{2}\|\omega\|_F^2 \tag{12}$$

quantifies the total rotational energy. Rising enstrophy indicates increasing systemic rotational stress.

4.8.3 BKM Blow-Up Criterion

The Beale–Kato–Majda blow-up criterion from fluid dynamics theory is adapted as a financial singularity indicator:

$$\int_0^T \|\omega(t)\|_\infty \, dt < \infty \tag{13}$$

If this integral diverges, the system is approaching a financial singularity analogous to turbulence breakdown in fluid mechanics.

4.8.4 Navier–Stokes BSDT Channels

Four composite channels merge the fluid dynamics variables with the blind-spot decomposition:

1. δ_C: Enstrophy anomaly (vorticity-based camouflage)
2. δ_G: Spectral gap (strain-based feature gap)
3. δ_A: Alignment anomaly (flow alignment deviation)
4. δ_T: Temporal novelty (flow pattern novelty)

4.8.5 Adaptive Viscosity

The adaptive friction coefficient is reinterpreted as an adaptive viscosity in the Navier–Stokes analogy:

$$\nu(E_{\mathrm{BS}}) = \nu_0 \left(1 + \gamma^*(E_{\mathrm{BS}})\right) \tag{14}$$

where ν_0 is a baseline viscosity. Higher blind-spot energy increases the viscosity (regulatory friction), damping dangerous vortical flows.

4.8.6 Minsky Regime Classification

The system classifies the financial regime into three Minsky stages based on the cosine alignment $\cos\theta$ between the gravitational gradient and the blind-spot gradient:

- **Hedge regime:** Gradients well-aligned ($\cos\theta > \tau_H$); standard risk measures capture the dominant sources of instability.
- **Speculative regime:** Gradients moderately misaligned; hidden risk building but not yet critical.
- **Ponzi regime:** Gradients strongly anti-aligned ($\cos\theta < \tau_P$); systemic risk is dominated by blind-spot energy invisible to equilibrium-based measures.

4.9. Online Algorithm and Computational Complexity

The system operates as an online algorithm within an Online Convex Optimisation (OCO) framework:

- **Per-step complexity:** $O(N^2 d \log N)$ dominated by the power iteration for spectral radius computation.
- **Regret bound:** $O(\sqrt{T})$ cumulative regret relative to the best fixed policy in hindsight.
- **Memory:** $O(N^2 d)$ for the interaction matrix and state.

4.10. Data Sources and Reference Period

In the preferred embodiment:

- **FDIC SDI call reports:** Quarterly financial data for US commercial banks, 1994–2028 (140+ quarters).
- **Reference (normal) period:** 1994-Q1 to 2003-Q4 (40 quarters), pre-dating the Global Financial Crisis.
- **Covariance estimation:** Ledoit–Wolf Oracle shrinkage estimator applied to inter-institution leverage correlations.
- **Institution types:** Seven categories (large commercial, community, savings, credit unions, investment banks, insurance, GSEs) in the aggregated model; 30 individual banks in the bank-level model; 20 global systemically important banks (G-SIBs) in the cross-border model.

4.11. Performance Characteristics

Configuration	N	T	**AUROC**	**GFC Lead**
Institution-type (aggregated)	7	140	≥ 0.70	6Q
Bank-level (individual)	30	140	0.805	6Q
G-SIB cross-border	20	76	0.781	6Q

Finding	**Value**
Super-criticality prevalence	68.6% of sample above $\mathcal{C}$
Out-of-sample GFC alarm	2007-Q1 (6 quarters before Lehman)
GFC welfare loss of inaction	19.57 pp consumption-equivalent
Peak control intensity (optional CCyB mapping)	231 basis points
Herding discovery	$\beta_{\delta_T} = -1.38$ (negative temporal novelty)
Scale invariance	Exact ($\rho = 2.41$ across scales 10^{-3}–10^{3})
Adversarial robustness	5 of 12 attack types never break detection

4.12. Robustness Verification

The system includes three robustness checks:

1. **Alternate normal period:** Shifting the reference period by ±4 quarters.
2. **Rolling-window refit:** Refitting reference distributions on a rolling basis.
3. **Leave-one-crisis-out:** Excluding each crisis in turn and testing detection of the excluded crisis.

Additionally, 12 adversarial attacks are tested, including bank removal, coordinated shift, gradient attack, noise injection, and data poisoning. Five of twelve attack types never break detection across all configurations.

5. CLAIMS

The following claims define the scope of protection sought. Where a claim refers to another claim, the dependency is by claim number. Features of any dependent claim may be combined with any independent claim or with other dependent claims.

1. A computer-implemented system for monitoring and controlling systemic risk in a financial network comprising a plurality of financial institutions, the system comprising:

   (a) a gravitational interaction engine configured to model the plurality of financial institutions as particles in a multi-dimensional feature space interacting through a pairwise potential comprising a bounded attractive component and a repulsive component;

   (b) a spectral monitoring module configured to compute the spectral radius of an interaction matrix derived from the financial network state, and to determine whether the financial network is above or below a critical manifold defined by the condition that the spectral radius exceeds a threshold;

   (c) a blind-spot energy module configured to compute a complementary energy function from reference distributions fitted on a normal reference period; and

   (d) an adaptive policy module configured to compute a state-dependent friction coefficient from the combined systemic energy and its gradient, and to map the friction coefficient to a transaction-flow control configuration for a transaction processing network, the configuration comprising:

      (i) a risk threshold profile defining numerical thresholds for transaction approval decisions; and

      (ii) a proof-requirement policy defining, for at least one threshold tier or transaction class, one or more cryptographic proofs required to be verified as a prerequisite to settlement.

2. A computer-implemented method for computing a state-adaptive transaction-flow control action for a financial transaction processing network, the method comprising:

   (a) receiving financial metric data for a plurality of financial institutions at a current time step;

(b) computing a state matrix $X \in \mathbb{R}^{N\times d}$ from the received data, where N is the number of institutions and d is the number of features;

(c) computing a pairwise interaction potential comprising an erf-attraction term and a logarithmic repulsion term for each pair of institutions;

(d) computing the spectral radius $\lambda_{\max}$ of an interaction matrix;

(e) determining whether the state matrix lies above or below a critical manifold $\mathcal{C} = \{X : \lambda_{\max}(\rho(X) \cdot \ell(X) \cdot W) = 1\}$;

(f) computing a blind-spot energy $E_{\text{BS}}(X)$ from a Mahalanobis distance to reference distributions fitted on a normal period;

(g) computing an optimal adaptive friction coefficient:

$$\gamma^*(X) \approx \frac{\beta\|\nabla E\|^2}{2\kappa + \beta\|\nabla E\|^2} \cdot \frac{E(X)}{E(X)+\theta}$$

where E is the combined energy, κ is an economic cost parameter, β is a systemic damage coefficient, and $\theta = \kappa/\beta$; and

(h) mapping the adaptive friction coefficient to a risk threshold profile defining numerical thresholds for transaction approval decisions; and

(i) mapping the adaptive friction coefficient to a proof-requirement policy defining, for at least one threshold tier or transaction class, one or more cryptographic proofs required to be verified as a prerequisite to settlement.

3. A system according to claim 1, wherein the pairwise potential is defined as:

$$\phi(r) = \gamma \cdot \frac{\sigma\sqrt{\pi}}{2} \cdot \operatorname{erf}(r/\sigma) - \gamma\lambda_{\text{rep}} \cdot \log(r+\varepsilon)$$

where r is the inter-institution distance, γ is an interaction strength, σ is an interaction scale, λ_{rep} is a repulsion strength, and ε is a regularisation constant.

4. A method according to claim 2, further comprising the step of computing the welfare cost of not implementing adaptive friction as:

$$\%CL = 100 \times \left(1 - e^{-\mathcal{L}_{\text{inaction}}/(\sigma\theta T)}\right)$$

where $\mathcal{L}_{\text{inaction}}$ is the accumulated loss under no intervention.

5. A method according to claim 2, wherein at least one further transaction-flow control action comprises applying a transaction rate limit, applying a queue delay, setting an escrow duration, or setting a maximum transaction value limit.

6. A method according to claim 2, wherein a further output comprises a countercyclical capital buffer adjustment computed as:

$$\Delta\text{CCyB}(t) \approx \kappa^{-1} \cdot \gamma^*(X_t) \cdot \sigma_\ell(t)$$

where $\sigma_\ell(t)$ is the system leverage volatility.

7. A system or method according to any preceding claim, further comprising a Navier–Stokes analogy module configured to:

(i) compute a velocity gradient tensor from changes in institution positions over successive time steps;

(ii) decompose the velocity gradient tensor into symmetric (strain rate) and anti-symmetric (vorticity) components;

(iii) compute enstrophy as $\mathcal{E} = \frac{1}{2}\|\omega\|_F^2$ from the vorticity tensor ω; and

(iv) evaluate a Beale–Kato–Majda blow-up criterion for detection of financial singularities.

8. A system or method according to any preceding claim, wherein the adaptive friction coefficient is reinterpreted as an adaptive viscosity $\nu = \nu_0(1+\gamma^*(E_{\mathrm{BS}}))$ in the Navier–Stokes analogy.

9. A system or method according to any preceding claim, further comprising a Minsky regime classification module configured to classify the financial network state into hedge, speculative, and Ponzi regimes based on the cosine alignment between the gradient of the gravitational potential and the gradient of the blind-spot energy.

10. A system or method according to any preceding claim, wherein the spectral radius is computed using power iteration with finite-difference Hessian-vector products, achieving $O(N^2 d \log N)$ per-step complexity.

11. A system or method according to any preceding claim, wherein the blind-spot energy comprises four deviation channels: an enstrophy anomaly channel, a spectral gap channel, an alignment anomaly channel, and a temporal novelty channel, each computed relative to reference distributions fitted on a designated normal period.

12. A system or method according to claim 1 or claim 2, further comprising a Lyapunov equivalence verification module configured to verify that the blind-spot energy and the gravitational potential are equivalent Lyapunov functions satisfying $c_1\|\nabla\Phi\| \leq \|\nabla E_{\mathrm{BS}}\| \leq c_2\|\nabla\Phi\|$ for positive constants c_1, c_2.

13. A method according to any preceding claim, wherein the method operates as an online algorithm within an online convex optimisation framework with $O(\sqrt{T})$ cumulative regret relative to the best fixed policy in hindsight.

14. A system or method according to any preceding claim, wherein the financial metric data comprises quarterly regulatory filings from a financial regulatory authority, and the reference period comprises at least 20 consecutive quarters of pre-crisis data.

6. DRAWINGS

[To be filed separately: Schematic of the GravityEngine showing N-body interactions; plot of the erf-log pairwise potential function; diagram of the critical manifold $\mathcal{C}$ in state space; flow diagram of the adaptive policy computation; time series showing the GFC early warning signal at 2007-Q1; Navier–Stokes analogy schematic showing velocity field, vorticity, and enstrophy; Minsky regime classification diagram.]

ABSTRACT

A computer-implemented system monitors and controls systemic risk in financial networks by modelling the financial system as a multi-body dynamical system. A gravitational interaction engine models financial institutions as particles with erf-attraction and logarithmic repulsion. A spectral monitoring module computes the spectral radius of the interaction network to detect phase transitions toward a critical manifold. A blind-spot energy module computes a complementary Lyapunov function from a four-channel failure mode decomposition. An adaptive policy module derives a state-dependent friction coefficient from the Bellman equation; the coefficient is mapped to transaction-flow control comprising a dynamically selected risk threshold profile and a proof-requirement policy gating settlement. The system is validated on 140 quarters of FDIC regulatory data, detecting the Global Financial Crisis six quarters in advance without using crisis labels during training.